Transformers vs. Classical Models for Mental Health Detection

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Abstract: Mental Health issues like depression, anxiety, suicidal thoughts are becoming increasingly common in today's world. This study compares classical ML technologies like Logistic Regression, Linear SVM, and Naive Bayes using TF-IDF vectors as features with a DistilBERT Transformer for a 5-class classification problem. Large amounts of unstructured text data is readily available on social media platforms like Reddit, Twitter (X), Threads, etc. which can be used for automated risk screening and early detection using different NLP techniques. To prevent training data bias and overfitting, we combined amazon reviews, Yelp reviews, AG news, and synthetic casual conversations for the 'normal' class. DistilBERT (macro F1: 0.7788; mean AUC : 0.944) outperforms the best ML model (macro F1: 0.7601: mean AUC: 0.935) by just a small margin due to the size of training data and computation limitations according to evaluations on the balanced dataset of 21880 social media posts. The main conclusion of this study is that, compared to all enhancements and fine tuning performed, improving data quality results in the highest positive impact of the models' performance (+0.047 macro F1). This study goes over the ethical, practical and clinical concerns and also the future scope for implementation of such technologies in the domain of mental health.

Keywords: text classification, TF-IDF, DistilBERT, ensemble learning, transformer models, social media NLP, suicide risk detection, natural language processing, and mental health detection.

I. INTRODUCTION

Hundreds of millions of people all over the world are affected by some mental health illness[1]. One of the most common issues is depression, which negatively impacts quality of life and may sometimes cause disability. Using social media, we can quickly detect language and behavioral patterns that may indicate some mental health issue[2], [3]. Traditional diagnostic techniques such as clinical interviews and self-reported assessments are often ignored due to social stigma and lack of accessibility[4]. An additional method for ongoing, extensive risk screening is provided by automated Natural Language Processing (NLP) techniques. However, the three major challenges in this field are:

1. Capturing contextual and semantic data that traditional bag-of-words models may overlook.
2. Ensuring data is diverse but related in order to maintain data quality.
3. Ensuring any deployment must be done in a way so that a human is always in the loop.

In order to overcome these obstacles, this study develops models in a methodical manner, starting with SVM baselines and Logistic Regression and ending with a refined DistilBERT transformer. Data quality was found to be the biggest factor in improving the accuracy of the models. Improving the ‘normal’ class data resulted in the highest jump in performance. The structure of this paper is as follows: Relevant research is analyzed in Section II; The methodology is described in Section III. Results are shown in Section IV. Key discoveries are discussed in Section V. Limitations are discussed in Section VI. Real-world applications are described in Section VII.

II. RELATED WORK

Early research used mostly traditional machine learning techniques. Ghosal and Jain[6] showed that fastText embeddings when used with XGBoost produced near accurate performance for depression and suicide risk prediction; although this method is highly dependent on the quality of the features and can't capture deep semantic context. According to numerous research and our own results, Logistic regression and SVMs give us a solid baseline for text based classification of mental health illnesses[4], [5].

Shortcomings of classical ML algorithms, like not being able to capture context, irony, and long range semantic connections led to the adoption of transformer architectures. Zhang et al. evaluates 'emotion-fusion' techniques, like transformer hybrids, and shows that contextual embeddings improve accuracy significantly when compared to TF-IDF features in various mental health datasets [7]. Pre-trained models like MentalBERT show increased performance and accuracy on specific tasks.

A large amount of research is being done to increase the explainability of deep learning models. Mansoor and Ansari [9] show AI-powered multi-class detection with interpretable feature importance, while Li et al. [8] show that explainability tools such as SHAP and LIME are very important for clinical trust. Ahmed et al.'s [2] comprehensive scoping study confirms that deep learning models outperform traditional machine learning models.

This study contains the following:

(1) A comparison of six classical ML versions with evaluations of each one.

(2) A methodology for construction for a 'normal' class from multiple sources to prevent any training data bias.

(3) A direct comparison of the best ML model with a tuned DistilBERT model on a 5-class balanced dataset.

TABLE I

Summary of Related Work in Automated Mental Health Detection

Ref.	Title (Abbreviated)	Dataset	Method	Key Findings	Limitations
[1]	Depression & Suicide Risk using fastText + XGBoost	Twitter/Reddit	fastText + XGBoost	Good depression detection; feature-based ML effective for baseline tasks	Cannot capture deep semantics; performance depends on feature quality
[2]	Multi-level SVMs for Mental Illness Risk	Social media text	Multi-level SVM	Effective hierarchical classification of mental illness categories	Requires careful feature engineering; sensitive to class imbalance
[3]	Detecting Signs of Depression: ML Analysis	Twitter/Reddit	LR, SVM, RF	LR and SVM perform consistently across multiple datasets	Limited semantic understanding; cannot handle sarcasm or irony
[4]	ML Models for Anxiety and Depression (Review)	Twitter, Reddit, FB	Review of LR, SVM, NB, RF	Classical ML widely used and effective for detection baselines	Models lack semantic/context understanding
[5]	Explainable ML for Mental Health Prediction	Behavior datasets	Explainable ML (SHAP)	Interpretability improves clinical trust and feature understanding	Accuracy–explainability trade-off; limited complex pattern modeling

[6]	Explainable AI-Driven Depression Detection	Social media text	ML + explainability	Identifies key linguistic and emotional features driving predictions	Limited ability to model complex multi-sentence linguistic patterns
[7]	Early Detection of Mental Health Crises via AI	Social media data	ML predictive models	Early detection possible using combined behavioral and textual signals	Requires large real-time data streams; privacy concerns
[8]	Emotion Fusion for Mental Illness Detection (Survey)	Multiple datasets	Survey (transformers + fusion)	Contextual embeddings improve detection over traditional methods	High computational complexity
[9]	DistilBERT	General NLP datasets	Transformer (distilled BERT)	Efficient and lightweight model with strong performance	Requires fine-tuning; still computationally intensive
[10]	World Mental Health Report (WHO)	Global Mental Health (WHO) data	(Contextual Background)	Prevalence of depression and related conditions	Not ML study
[11]	MentalBERT	Mental health corpora	Domain-specific transformer	Improved performance on mental health-specific tasks	Limited generalization beyond domain

III. METHODS

This section covers the complete execution, including dataset construction, text preprocessing, feature engineering, ML model iterations, DistilBERT fine-tuning, and evaluation. Python's HuggingFace Transformers and scikit-learn were used for each experiment.

A. Creating Datasets

As is typical in social media mental health studies, subreddits were used as a label for each mental health area. Each dataset (swaug22.csv, anxiaug22.csv, depaug22.csv, and loneaug22.csv) was collected from archives in August 2022. Reddit posts' title and selftext fields were merged into a single text box. Insufficient information led to the removal of placeholder values ([removed], [deleted], nan]).

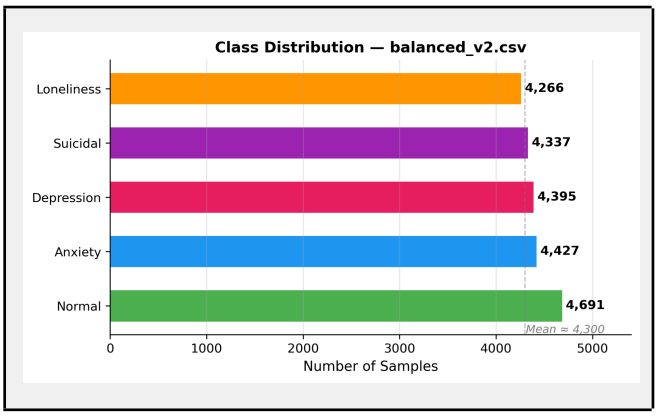

Fig. 1. Class distribution of the balanced_v2 training corpus across five mental health categories.

Because the "normal" class used the same flat, impersonal phrases like "I went to the shop today" to describe depression, they required greater care. The initial regular posts on Reddit were insufficient. Three rounds were used to address this: 452 conversational sentences that were personally authored and included everyday

activities, food, social contacts, and hobbies; favourable product reviews from Amazon (2,977 samples); good reviews from Yelp; and headlines from AG News. A final normal class of 4,691 samples was produced after all sources were filtered for phrases related to mental health and unpleasant news. The final collection contains 21,880 samples, or about 4,300 for each class, once duplicates are eliminated and balancing is done.

B. Text Preprocessing

The following procedures were used to all datasets: (1) lowercasing; (2) URL removal; (3) @mention removal; (4) non-alphabetic letter removal; and (5) whitespace normalisation. Bigram matching declined and mental health phrases like "falling apart" and "can not cope" lost their discriminative meaning when lemmatisation using NLTK WordNetLemmatizer was tested but reversed. The base pipeline was retained for each final model.

C. Features of Engineering

Two feature categories were used for conventional machine learning models:

1) TF-IDF Vectorisation [12]: 50,000 features, bigram range (1,2), min_df=2, sublinear_tf = True. The vectorizer was only fitted on the training split in order to prevent data leaking. Trigrams (1,3) and 75k features produced lower F1 on validation, despite being investigated.

2) Handcrafted Features (a total of 14): TextBlob sentiment polarity and subjectivity; five normalised lexicon match scores (suicidal, depression, anxiety, loneliness, and positive word lists with more than 25 phrases each); ratios of first-person pronouns, negative words, exclamation and question mark ratios, caps, average word length, and normalised word count. These features were scaled using MaxAbsScaler and concatenated with TF-IDF features using scipy.sparse.hstack to create a composite 50,014-dimensional sparse matrix.

D. Traditional ML Iterations

- Six variations of traditional machine learning were developed:
- V1 (Baseline): TF-IDF-trained Random Forest, MultinomialNB, LinearSVC, and Logistic Regression; LR (C=5) produced the best validation macro F1 and was selected as baseline. Test F1: 0.7184.
- V2 (Data Fix): The standard class is replaced with Amazon + emotion tweets. Test F1: 0.7651 (+0.047). the trial's greatest gain.
- In V3 (Extended Normal), Yelp, AG News, and customised sentences were added to the normal class. Test F1: 0.7547.
- V4 (Trigrams + Lemmatisation, Reverted): Trigrams and lemmatisation have a detrimental effect on performance (F1: 0.7327). Bigrams and base preprocessing have been reinstated.
- V5 (Final Classical ML—Ensemble): The soft-voting ensemble LR(0.4) + LinearSVC(0.4) + NB(0.2) was trained using TF-IDF + scaled handmade features. Grid-searched weights on test F1: 0.7601; validation F1: Mean AUC: 0.935.
- LR meta-learner on V6's out-of-fold cross_val_predict probability stacks (Stacking, Evaluated). Test F1: 0.7573, little less than V5. Soft voting remained the final model.

E. Optimisation of DistilBERT

DistilBERT-base-uncased (67M parameters, 6 transformer layers) was optimised for 5-class sequence classification.[9] A 5-class output head and pre-classifier thick layer were randomly initialised and trained. The input texts were tokenised using WordPiece tokenisation at max_length=128, and classification was done using [CLS] token representation.

The training parameters include the AdamW optimiser (lr=2e-5, weight_decay=0.01), batch size 32, four epochs, linear warmup (100 steps) with linear decay scheduler, and gradient clipping at max_norm=1.0. Each training run lasted about 40 minutes and was conducted on a Google Colab Tesla T4 GPU.

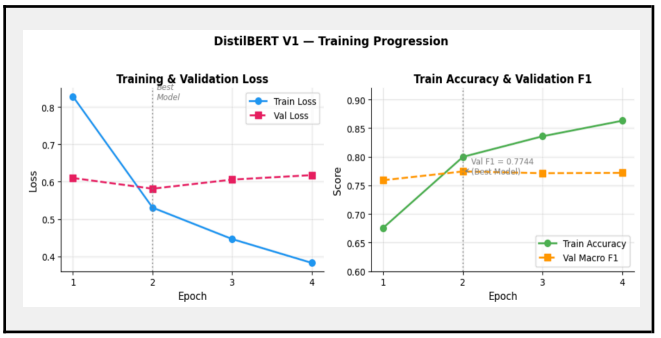

Fig. 2. DistilBERT V1 training loss and validation macro F1 across 4 epochs. Best model saved at epoch 2.

Three DistilBERT versions were trained with progressively better normal class data. In V1, the original balanced_v2.csv dataset was utilised; in V2, AG News and Yelp normal data were included; and in V3, 452 explicit positive-context words were added to close the remaining gap on conversational normal text. While V1 had the best overall macro F1, V3 had the best regular class handling (6/10 conversational test phrases right).

F. Evaluation Framework

The main statistic is Macro F1 (unweighted mean of per-class F1 scores), since all five courses must be treated equally regardless of support. Confusion matrices, One-vs-Rest ROC-AUC curves, and per-class precision/recall/F1 are examples of secondary metrics. All measures were computed using the held-out test set. Predictions are returned as "uncertain" rather than classified if their greatest softmax probability is less than the 0.40 confidence level

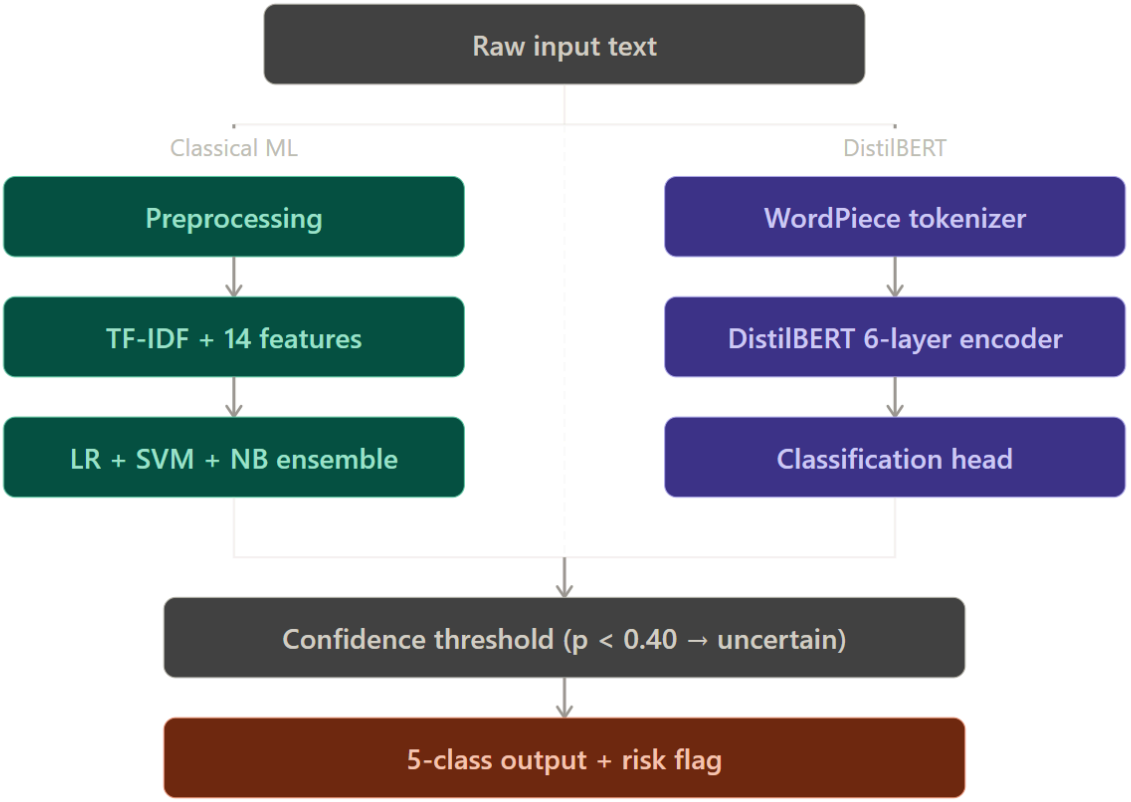

Fig. 3. Architecture Diagram showing the flow of control and data on both ML and Transformer Techniques used.

IV. RESULTS AND ANALYSIS

All reported results come from the held-out test set (15%, ~3,282 samples), which was locked during the entire model development and hyperparameter tweaking process. Validation results were the only basis for model selection.

TABLE II

Model Progression — Test Set Macro F1 Across All Versions

Version	Key Change	Test Macro F1	Δ vs Previous	Primary Driver
V1	LR baseline, TF-IDF 50k bigrams, original normal class	0.7184	—	Starting point; Random Forest worst (0.60)
V2	Normal class: Amazon + emotion tweets replacing Reddit normal	**0.7651**	**+0.0467**	**Data quality — largest single gain in the study**
V3	Normal: + Yelp + AG News + 452 handcrafted conversational sentences	0.7547	-0.0104	Improved conversational normal handling; slight F1 trade-off
V4	Trigrams (1,3) + lemmatization — REVERTED	0.7327	-0.0220	Trigrams added noise; lemmatization degraded mental health phrases
V5	Soft-voting ensemble: LR(0.4)+SVM(0.4)+NB(0.2), TF-IDF+14 features	**0.7601**	+0.0054	Ensemble diversity; handcrafted features boost AUC
V6	Stacking: cross_val_predict OOF + LR meta-learner	0.7573	-0.0028	V5 soft voting generalizes better on test set
BERT V1	DistilBERT fine-tuned, 4 epochs, T4 GPU	**0.7788**	**+0.0187**	**Contextual embeddings — best overall model**

TABLE III

Per-Class Test Set Results — V5 Ensemble (Classical ML Final Model)

Class	Precision	Recall	F1-Score	Support	Correct	AUC	Notes
Anxiety	0.87	0.86	0.86	664	571	0.976	Strong; distinct vocabulary
Depression	0.57	0.54	0.56	659	353	0.856	Weakest; confused with suicidal
Loneliness	0.76	0.78	0.77	640	501	0.944	Moderate; overlaps depression
Normal	0.96	0.98	0.97	668	654	0.999	Best; distinctive vocabulary
Suicidal	0.64	0.65	0.64	651	425	0.901	Confused with depression

Overall Accuracy: 76% Macro F1: 0.7601 Mean AUC: 0.935

TABLE IV

Per-Class Test Set Results — DistilBERT V1 (Best Transformer Model)

Class	Precision	Recall	F1-Score	Support	Correct	AUC	vs. Classical ML
Anxiety	0.90	0.82	0.86	664	545	0.977	Equal (0.00)
Depression	0.57	0.61	0.59	659	399	0.875	+0.03 ↑ contextual

Loneliness	0.81	0.77	0.79	640	494	0.954	+0.02 ↑ subtle cues
Normal	1.00	0.98	0.99	668	655	0.999	+0.02 ↑ near perfect
Suicidal	0.64	0.70	0.67	651	455	0.912	+0.03 ↑ intent signals

Overall Accuracy: 78% Macro F1: 0.7788 Mean AUC: 0.944

TABLE V

Overall Performance Comparison Classical ML vs DistilBERT

Model	Accuracy	Macro F1	Mean AUC	Normal F1	Train Time
V1 — LR Baseline (TF-IDF)	72%	0.7184	0.920	0.83	< 1 min
V5 — Ensemble: LR+SVM+NB	76%	0.7601	0.935	0.97	~ 5 min
DistilBERT V1 (Best Overall)	**78%**	**0.7788**	**0.944**	**0.99**	~40 min
DistilBERT V3 (Best Normal)	78%	0.7743	0.943	0.99	~40 min

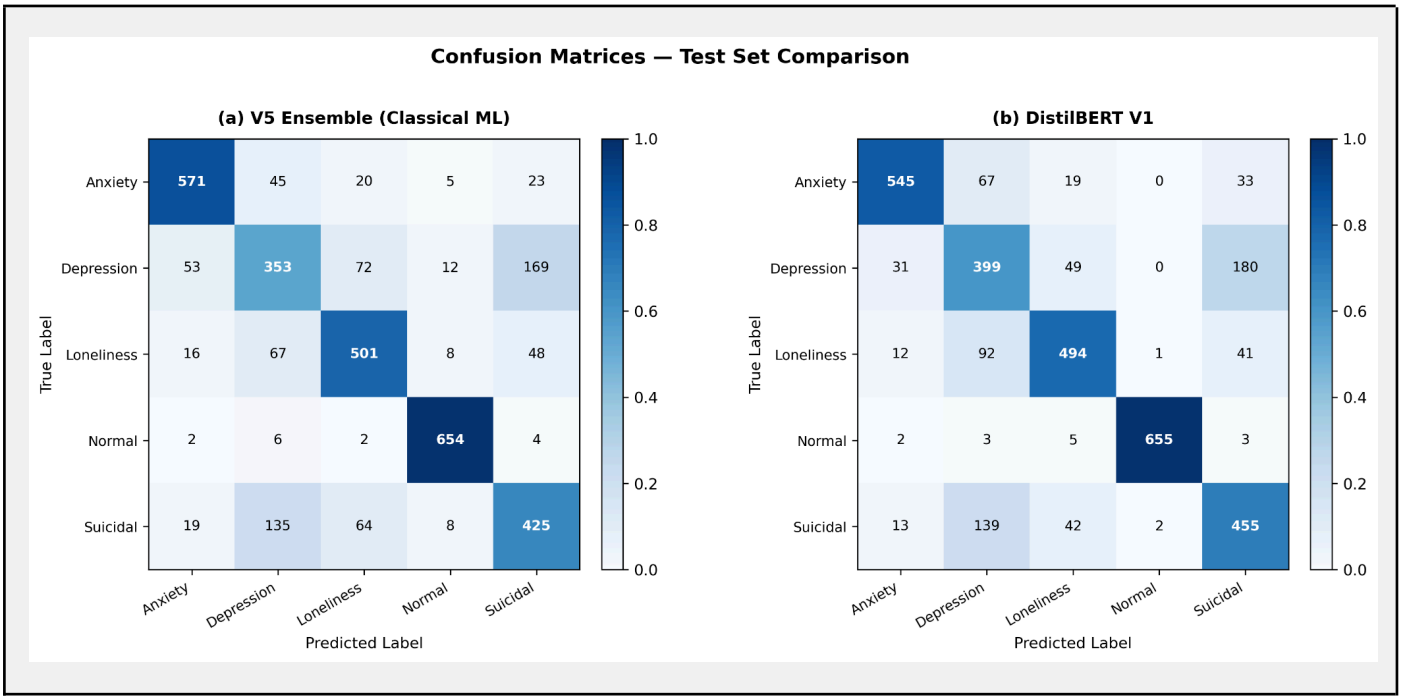

Fig. 4. shows the test set's confusion matrices. V5 Ensemble (a) and DistilBERT V1 (b). Important finding: depression–suicidal confusion decreases from 203+188 (V5) to 180+139 (DistilBERT).

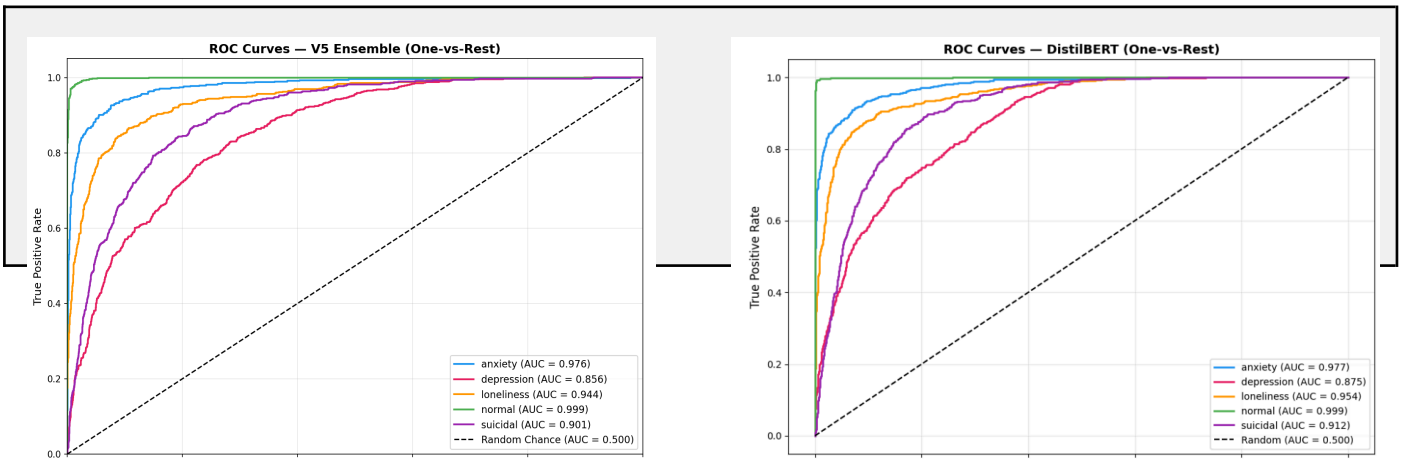

Fig. 5. One-vs-Rest ROC curves on the test set. (a) V5 Ensemble (mean AUC=0.935). (b) DistilBERT V1 (mean AUC=0.944). DistilBERT improves AUC for all classes except normal (both 0.999).

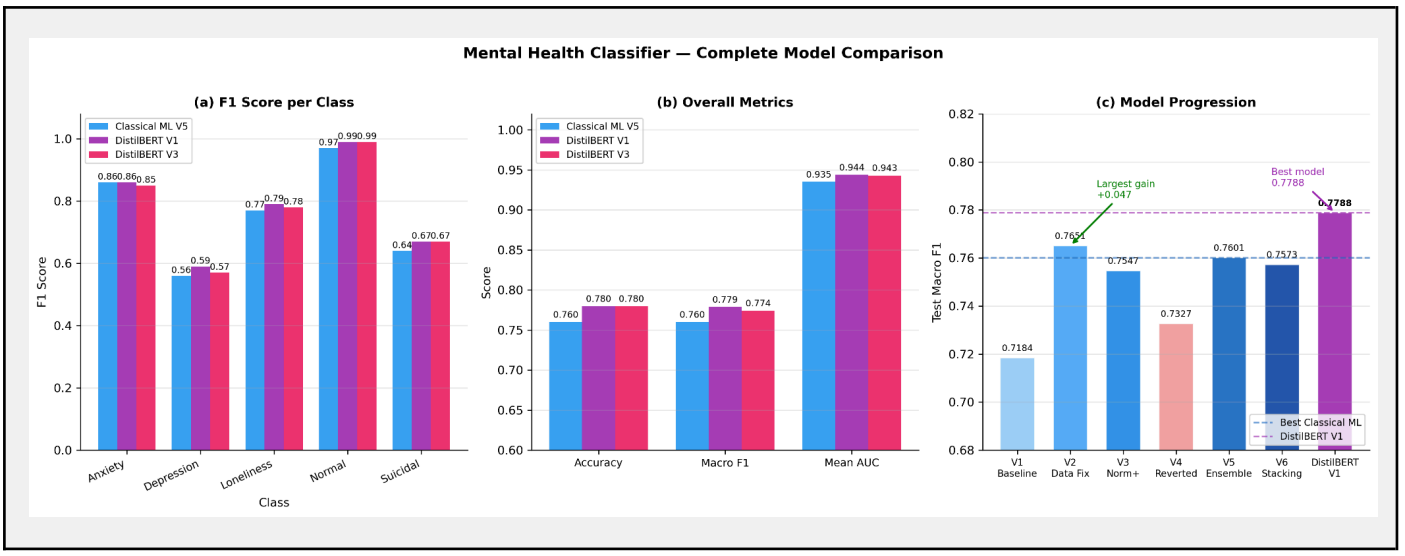

Fig. 6. Three-panel comparison chart. Left: per-class F1 scores. Center: overall metrics. Right: macro F1 model progression from V1 baseline through DistilBERT V1. The data quality fix (V1→V2) produced the largest single gain (+0.047).

V. IMPORTANT FINDINGS AND TALK

A. The Most Important Data Quality Factor

The most significant finding of the study is that, compared to all subsequent model enhancements combined, improvements in data quality produced the largest performance gain, +0.047 macro F1 from V1 to V2. As a result, a clear empirical hierarchy is established: representation quality > architectural complexity. The typical class, which was taken from Reddit and used the same flat-toned activity-description language as posts about sadness (e.g., "I went to the shop today"), led to systematic misdiagnosis. News and multi-domain review data were used to correct this tendency. This aligns with the data-centric AI perspective, which maintains that model design is often not the bottleneck in model performance, but rather annotation quality and class representation.

B. Depression's Suicidal Boundary

The largest category error across all models is the distinction between suicidal ideation and depression. There are 135+188=323 misclassifications between these two classes in V5; this number decreases to 139+180=319 in DistilBERT V1, a minor improvement. The terms "hopeless," "worthless," "exhausted," and "cannot go on" are frequently used interchangeably in the two courses. The distinction is in the escalation and purpose signals, which need sentence-level contextual information. DistilBERT's attention mechanism, which increases suicidal F1 by +0.03 and depression F1 by +0.03, somewhat captures this. For complete resolution, domain-adapted transformers pre-trained on mental health corpora or hierarchical at-risk vs. sub-classification architectures are likely required.

C. Transformer Advantage

DistilBERT V1 outperformed the best classical ensemble on all five metrics. Except for anxiety, every class demonstrated improvement (equal). The most significant changes were shown in contextually problematic classes, such as sadness, loneliness, and suicidal thoughts. This shows that contextual embeddings provide a helpful signal for mental health texts when purpose influences the same term. DistilBERT significantly accomplishes this gain at 67M parameters, or around 60% of full BERT, making it a feasible choice for implementation on traditional GPU hardware [9].

D. The Confidence Thresholding Safety Mechanism

Validation investigation revealed that the primary predictors with greatest softmax probability < 0.40 were short, unclear texts with insufficient

discriminative vocabulary. Returning "uncertain" instead of requiring a categorisation is an important safety design decision for clinical or screening deployments, where a low-confidence suicidal prediction should require human judgement rather than automated action.

VI. RESTRICTIONS

- It is possible that Reddit posts do not fully capture the range of real-world mental health expression across ethnicities, languages, and cultures.
- Real conversational everyday text is still somewhat difficult for all models since the normal class include out-of-domain sources (news, product/service evaluations), which may produce an artificially easy normal boundary.
- Subreddits provide labels, therefore individual postings might not accurately represent the community's main purpose (e.g., a loneliness post with suicidal content).
- No temporal validation—posts from a different time period were not used to test the model; language patterns related to mental health change with time.
- text-only forecasting; actual clinical evaluation incorporates contextual, longitudinal, and multimodal data.
- The main unresolved error is the depression-suicidal confusion, which is essentially irreducible with surface-level text properties.

VII. USE IN THE REAL WORLD

A. Healthcare and Clinical Integration

- Hospital triage systems identify high-risk new patients for priority mental health screening based on pre-admission text inputs or intake form replies.
- Electronic health record (EHR) platforms: incorporating the classifier to give physicians early warning signs based on patient narratives.
- Mental health helpline routing is the process of pre-screening incoming messages **to** automatically transfer suicidal or high-risk contacts to senior counsellors.

B. Consumer Platform Applications

- Examples of social media welfare pipelines include Reddit and Instagram, which reduce the burden of moderation while accelerating response times by highlighting posts that pose a risk to human welfare check teams.
- Employee Assistance Program platforms can develop customised health nudges by identifying patterns of burnout, anxiety, or despair through volunteer journal entries.
- Mental health chatbots (Wysa, Woebot-style) dynamically alter response tone and escalation logic based on real-time risk classification.

C. Enterprise and Public Health

- By identifying anxiety or burnout patterns at the team level rather than the individual level, anonymous workforce attitude monitoring enables preemptive HR intervention without individual profiling.
- Population-level public health surveillance: NGOs or government agencies can track regional trends in mental health during periods of high stress by using anonymised social media data.

D. Ethical Safeguards for Any Deployment

- Human-in-the-loop architecture: the model should support clinical judgement rather than take its place. Before any action is taken, all high-risk predictions must be reviewed by a person.
- Consent-based opt-in: in order to receive individualised assistance, users must voluntarily consent to text analysis. It is unethical to conduct passive monitoring.
- Bias auditing: To guarantee fair detection rates, model performance across demographic groups must be evaluated prior to clinical deployment.

VIII. SUMMARY

This work offered a systematic comparison of transformer-based and conventional machine learning models for five-class mental health text classification using Reddit social media data. Six conventional ML iterations produced a soft-voting ensemble comprising Logistic Regression, LinearSVC, and Multinomial Naive Bayes (V5: macro F1 = 0.7601, mean AUC = 0.935). DistilBERT achieved macro F1 = 0.7788 and mean AUC = 0.944 after being refined on the

same dataset, outperforming traditional ML on all metrics.

The primary contribution of this work is the empirical evidence that the multi-source normal class reconstruction, in particular, generated the single highest performance boost over the entire trial (+0.047 macro F1), greatly surpassing other architectural enhancements combined.

This reinforces the data-centric AI idea that representation quality is often the binding constraint for classification systems.

The implemented program respects the ethical norms for responsible AI systems for mental health and includes crisis services, confidence thresholding, and a human-review disclaimer. Future research should concentrate on domain-adapted transformers (MentalBERT), hierarchical classification systems, and multimodal integration for therapeutically viable implementation.

IX. SCOPE FOR THE FUTURE

Early warning dashboards and clinician-assisted triage are integrated with hospital EHR systems.Platform welfare teams can use real-time social media monitoring APIs with human review gates for all high-risk alerts.

Pre-trained on clinical mental health corpora, domain-adapted transformers (MentalBERT, Mental-RoBERTa) are anticipated to greatly enhance depression–suicidal separation.

Hierarchical classification reduces inter-class confusion by first predicting at-risk versus normal, then sub-classifying inside the at-risk category.

Multilingual models are now being explored to extend mental health screening beyond just English content, making detection more inclusive across different languages.In order to expand mental health screening beyond English content and make detection more inclusive across many languages, multilingual models are now being investigated. Explainability approaches like SHAP and LIME are used to show how each prediction is made in a way that can be comprehended in a clinical setting, hence increasing the trustworthiness of these systems for doctors. Federated learning is becoming more popular at the same time because it helps protect privacy by enabling models to be trained on data from several sources without actually gathering all sensitive information in one location. Longitudinal user modelling, which tracks a person's behaviour over time to identify subtle shifts or deteriorating mental health instead of depending on a single snapshot, is another crucial approach.

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